Spatial Reasoning in Large Language Models: A Comprehensive Analysis

Executive Summary

This document provides a comprehensive analysis of spatial reasoning capabilities in Large Language Models (LLMs), with specific focus on GPT-4o, Google Gemini, and Claude. The analysis evaluates current capabilities, limitations, and emerging approaches that bridge the gap between language understanding and spatial intelligence. Our findings reveal that while modern LLMs show impressive multimodal capabilities, they fundamentally lack true 3D spatial understanding and require integration with specialized vision-language-action systems for real-world deployment.

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1. Introduction: Defining the Problem Space

1.1 What is Spatial Reasoning in AI?

Spatial reasoning in artificial intelligence refers to the ability to understand, interpret, and manipulate spatial relationships between objects in both 2D and 3D environments. Unlike simple object recognition, spatial reasoning requires comprehension of:

Key Components:

- 2D vs 3D Reasoning: Understanding flat surfaces versus volumetric space
- Object Relationships: Distance, orientation, relative positioning, and occlusion
- Egocentric vs Allocentric Understanding:
  - Egocentric: Understanding space from one's own viewpoint
  - Allocentric: Understanding space from an external, objective perspective

1.2 Vision Understanding vs Spatial Reasoning

It is critical to distinguish between two different capabilities:

Vision Understanding (Recognition):

- Identifying objects in images
- Classifying visual elements
- Detecting patterns and features
- Image captioning and description

Spatial Reasoning (Understanding Relationships + Actions):

- Comprehending how objects relate to each other in space
- Predicting spatial consequences of actions
- Understanding depth, occlusion, and 3D structure
- Guiding physical actions based on spatial understanding

Research indicates that current LLMs excel at vision understanding but struggle significantly with true spatial reasoning tasks.

2. Current LLM Capabilities Evaluation

2.1 GPT-4o (OpenAI)

Multimodal Capabilities:
GPT-4o represents OpenAI's most advanced multimodal model with native integration of text, image, and video inputs. Recent updates have extended its training data cutoff from October 2023 to June 2024, providing more current knowledge.
Spatial Reasoning Performance:
According to recent benchmark studies, GPT-4o demonstrates mixed spatial reasoning capabilities:

- **Strengths:** Improved visual understanding and analysis of image uploads, better interpretation of spatial relationships compared to GPT-4 Vision
- **Documented Weaknesses:** Spatial reasoning accuracy of only 27.5% on viewpoint-based questions in specialized benchmarks
- **Performance Issues:** Struggles with precise localization, pattern recognition, analogy reasoning, and spatial movement calculations

Key Research Finding:
A 2024 empirical analysis found that GPT-4o, when tested on spatial reasoning tasks, showed significant limitations. The model was evaluated alongside Gemini 1.5 Pro and other LMMs on the Spatial-MM benchmark, revealing that perspective-based spatial understanding remains challenging.
Enhanced Capabilities with Visual Prompting:
Research on 3DAxisPrompt demonstrates that GPT-4o's 3D spatial grounding can be significantly improved by overlaying visible 3D axes, markers, and region edges on observation images. This visual augmentation enables better localization and spatial reasoning tasks.
Real-World Application Examples:

- Scene understanding for layout analysis
- Document spatial organization (improved with visual markers)
- Chart and graph interpretation
- Basic navigation assistance (limited accuracy)

2.2 Gemini (Google DeepMind)

Native Multimodality:
Gemini represents Google's integrated approach to multimodal AI, with spatial understanding as a core design principle rather than an added feature.
Spatial Reasoning Advancements:
Gemini 3 Pro (Latest Generation - 18 November 2025):

- State-of-the-art performance on spatial reasoning benchmarks
- No dedicated spatial-specific benchmark data publicly available — performance based on documented capabilities from official Google DeepMind sources
- Advanced capabilities in visual recognition, spatial understanding, and embodied reasoning

Gemini 2.5 Pro (Released: March 25, 2025 — Experimental | General Availability: June 17, 2025):

- Significant leap in spatial reasoning and math performance
- Enhanced sampling patience driving standout geometry performance
- Strong results across coding, math, and multimodal benchmarks

Specialized Robotics Models:
Google has developed dedicated spatial reasoning variants:
Gemini Robotics-ER (Embodied Reasoning):

- Advanced spatial understanding for robotics applications
- Enhanced 3D detection and pointing capabilities

- Real-time trajectory prediction and task progression
- Integration with low-level robot controllers

Key Capabilities:

- Understanding complex spatial relationships
- Shape mapping and visual problem-solving
- High frame-rate video understanding (10+ FPS)
- Spatial reasoning combined with code generation

Benchmark Performance:

Gemini models demonstrate superior performance in:

- Document understanding beyond simple OCR
- Spatial awareness for autonomous vehicles and XR devices
- Screen understanding for computer use agents
- Long-context video synthesis across hours of footage

2.3 Claude (Anthropic)

Vision Capabilities:

Claude's approach to vision is fundamentally different from GPT-4o and Gemini. Claude is primarily a language model with visual perception integrated into its reasoning framework, rather than a true multimodal architecture.

Spatial Reasoning Limitations:

Official documentation from Anthropic explicitly acknowledges spatial reasoning as a known limitation:

Key Weaknesses:

- Limited spatial reasoning abilities
- Difficulty with precise localization tasks
- Struggles with layout understanding
- Cannot accurately read analog clock faces
- Difficulty describing exact positions (e.g., chess piece locations)

What Claude Does Well:

Despite spatial limitations, Claude excels at:

- Document analysis and interpretation
- Visual reasoning about concepts rather than precise spatial relationships
- Chart and graph analysis (interpretive, not measurement-based)
- Image context understanding for reasoning tasks

Architecture Approach:

Claude reverses the typical multimodal approach: it is a language model with visual perception integrated, rather than a vision model with language capabilities added. This means Claude excels at reasoning about what it sees, but not at precise spatial measurements or relationships.

Recent Improvements — Claude 4 Family (May 2025 — April 2026):

Claude Opus 4 and Claude Sonnet 4 were officially released on May 22, 2025, as the fourth generation of Anthropic's AI models, featuring hybrid reasoning and extended thinking capabilities.

Subsequent updates include:

- Claude Opus 4.1 (August 2025)
- Claude Haiku 4.5 (October 2025)
- Claude Opus 4.5 (November 2025)
- Claude Opus 4.6 (February 2026)
- Claude Opus 4.7 (April 2026) — Currently most capable generally available model

Vision Improvements in Claude Opus 4.7 (Latest):
- High-resolution image support added (2576px / 3.75MP)
- Improved low-level perception: pointing, measuring, counting
- Improved image localization and bounding-box detection

Despite these improvements, spatial reasoning remains a documented limitation compared to GPT-4o and Gemini, as officially acknowledged in Anthropic's API documentation.

3. Comparison of LLM Capabilities

3.1 Comprehensive Capability Matrix

Capability	GPT-4o	Gemini 3 Pro	Claude 4
Input Modalities	Image, Video, Text	Image, Video, Text, Audio	Image, Text
Native Multimodality	Enhanced	Native from design	Language-first with vision
Depth Understanding	Limited (improved with prompts)	Advanced (dedicated models)	Limited
3D Spatial Reasoning	Weak (27.5% on viewpoints)	Strong (specialized variants)	Explicitly limited
Object Position Inference	Moderate	Strong	Weak
Layout Understanding	Moderate	Strong	Weak
Real-time Video Analysis	Yes (improved frame rate)	Yes (10+ FPS)	No (discrete images only)
Action Guidance	Limited	Strong (via Robotics-ER)	Not applicable
Primary Strength	Reasoning + Web Knowledge	Spatial Understanding + Embodiment	Language Reasoning + Interpretation
Primary Weakness	Precise spatial localization	High computational cost	Spatial relationships
Best Use Case	General multimodal tasks	Robotics, spatial planning	Document analysis, reasoning

3.2 Spatial Understanding Depth Analysis

Spatial Task	GPT-4o	Gemini	Claude
Object Recognition	Strong	Strong	Strong
Navigation Planning	Moderate	Strong	Strong — Most Detailed
Object Interaction Guidance	Strong — Practical Steps	Moderate — Avoids Physical Tasks	Weak — Declines Guidance
Scene Reasoning	Moderate	Moderate	Weak — Context Confusion
Distance Estimation	Limited	Strong	Limited
Depth Perception	Limited	Strong	Limited
Occlusion Reasoning	Moderate	Strong	Limited
3D Scene Reconstruction	Limited	Strong	Limited
Persistent Spatial Memory	None	None	None
Temporal Spatial Tracking	Limited	Moderate	Limited

Note: Performance ratings are based on direct practical testing conducted across navigation, object interaction, scene reasoning, and spatial memory scenarios, supplemented by findings from the Multimodal Spatial Reasoning Survey (arXiv, November 2025 — arxiv.org/abs/2510.25760) and EarthSpatialBench (arXiv, February 2026 — arxiv.org/pdf/2602.15918).

4. Multimodal Understanding of Physical Environments

4.1 How LLMs Interpret Physical Spaces

Current LLMs process physical environments through several mechanisms:

Image Processing:

- Pixel-level analysis through vision transformers
- Feature extraction using pre-trained encoders (CLIP, DINOv2, SigLIP)
- Semantic segmentation of visual scenes

Video Stream Understanding:

- Frame-by-frame analysis at varying frequencies
- Temporal relationship detection between frames
- Motion pattern recognition (limited)

Sensor-Like Inputs:

- Simulated depth maps (not true 3D understanding)
- Annotated spatial markers (3DAxisPrompt approach)
- Object bounding boxes and segmentation masks

4.2 Critical Limitation: LLM ≠ Spatial Engine

Fundamental Insight:

LLMs work on representation, not true 3D understanding. They process:

- 2D projections of 3D scenes
- Learned patterns from training data
- Statistical correlations between visual features and spatial descriptions

They do NOT have:

- Internal 3D world models
- Physics simulation capabilities
- True depth perception
- Persistent spatial memory across interactions

4.3 Vision-Language Models (VLMs) and World Models

Emerging Paradigm:

The integration of vision and language in models like Gemini represents an important trend toward "world models" - systems that build internal representations of physical environments.

Components of Advanced VLMs:

1. Vision encoders (process images)
2. Language models (process instructions)
3. Cross-attention mechanisms (link vision and language)
4. Action decoders (output spatial actions)

World Model Concept:

World models aim to predict spatial states and consequences of actions, moving beyond static scene understanding toward dynamic environment simulation.

Section 5: Practical Scenario Testing

5.1 Navigation Scenarios

Test Case: "How do I move from this room to the door?"
Test Image Provided: Office hallway with glass doors, green EXIT sign, plants, and open floor space.

GPT-4o Performance:

- Correctly identified the glass door on the right side of the image
- Identified the green EXIT sign above the door
- Provided clear directional guidance — walk forward, angle slightly to the right
- Confirmed the path was clear with no obstacles
- Success Rate: Good

Gemini Performance:

- Correctly identified the glass door in the middle-right area
- Identified the green EXIT sign and recommended following it
- Provided step-by-step navigation path
- Confirmed the path was clear and straightforward
- Success Rate: Good

Claude Performance:

- Correctly identified the glass door straight ahead
- Provided the most detailed response — included specific landmarks such as the green pedestal table and wooden planter box
- Explicitly mentioned the green EXIT sign and indicated a right turn at the end
- Success Rate: Best — Most Detailed Response

Performance Summary — Navigation:

Model	Door Identified	Path Guidance	Landmarks Mentioned	Overall
GPT-4o	Yes	Clear	Partial	Good
Gemini	Yes	Clear	Partial	Good
Claude	Yes	Clear	Detailed	Best

5.2 Object Interaction Scenarios

Test Case: "Pick the cup behind the laptop"
Test Image Provided: Office desk with open laptop, red cup and saucer placed behind the laptop, notebooks and pencil on the right side.

GPT-4o Performance:

- Correctly identified both the laptop and the red cup
- Understood the spatial relationship — cup is behind the laptop
- Provided practical step-by-step physical instructions on how to reach and pick up the cup
- Warned about potential obstacles such as bumping the screen
- Success Rate: Best — Most Practical Guidance

Gemini Performance:

- Correctly identified both the laptop and the red cup
- Understood the spatial relationship — cup is behind the laptop
- Acknowledged the task but avoided providing physical manipulation guidance
- Deflected by stating the image is only a photograph
- Success Rate: Partial — Avoided Physical Guidance

Claude Performance:

- Correctly identified both the laptop and the red cup
- Understood the spatial relationship
- Declined to provide physical guidance, stating it is an AI without a physical body
- Response was honest but practically unhelpful for the task
- Success Rate: Low — Declined Physical Guidance

Performance Summary — Object Interaction:

Model	Objects Identified	Spatial Relationship	Physical Guidance	Overall
GPT-4o	Yes	Correct	Detailed Steps	Best
Gemini	Yes	Correct	Avoided	Partial
Claude	Yes	Correct	Declined	Low

5.3 Scene Reasoning Scenarios

Test Case: "What objects are blocking the path?"
Test Image Provided: Office desk with laptop, red cup, and notebooks (same image as Scenario 2).
GPT-4o Performance:

- Correctly analyzed the laptop image
- Identified the laptop and laptop screen as the primary blocking objects
- Correctly noted that notebooks and pencil on the right were not blocking the direct path to the cup
- Success Rate: Correct

Gemini Performance:

- Correctly analyzed the laptop image
- Identified the laptop as the main blocking object
- Suggested either moving the laptop aside or reaching around it
- Success Rate: Correct

Claude Performance:

- Incorrectly analyzed the wrong image — provided object analysis based on the hallway image from Scenario 1 instead of the laptop desk image
- Identified hallway objects such as the green pedestal table and office chair as non-blocking items
- This response demonstrated a significant spatial context error
- Success Rate: Incorrect — Wrong Image Analyzed

Performance Summary — Scene Reasoning:

Model	Correct Image Analyzed	Blocking Objects Identified	Accuracy	Overall
GPT-4o	Yes	Laptop + Screen	Correct	Good
Gemini	Yes	Laptop	Correct	Good
Claude	No — Wrong Image	Hallway objects	Incorrect	Fail

5.4 Spatial Memory Scenarios

Test Prompt: "Where was the exit earlier?"
Test Setup:

- Image 1: Office desk with laptop and red cup (Scenarios 2 and 3)
- Image 2: Office hallway with glass doors and green EXIT sign (Scenario 1)
- Objective: To evaluate whether models can retain and recall spatial information from previously viewed images within a conversation

GPT-4o Response:

- Correctly identified the exit on the right side of the hallway
- Identified the green EXIT sign above the glass door
- Result: ✅ Correct

Gemini Response:

- Correctly identified the exit in the upper right corner
- Identified the green EXIT sign above the glass door
- Result: ✅ Correct

Claude Response:

- Correctly identified the exit in the background towards the right
- Identified the illuminated green EXIT sign hanging from the ceiling
- Result: ✅ Correct

Performance Summary — Spatial Memory:

Model	Exit Location	Sign Identified	Apparent Result	Actual Reason
GPT-4o	✅ Correct	✅ Yes	Passed	Context window only
Gemini	✅ Correct	✅ Yes	Passed	Context window only
Claude	✅ Correct	✅ Yes	Passed	Context window only

Critical Finding:
All three models — GPT-4o, Gemini, and Claude — appeared to successfully recall the exit location from the previously viewed hallway image. However, this does not constitute true spatial memory. The original image remained accessible within the same conversation context window, allowing each model to reference it directly when answering the question.
A genuine spatial memory test would require the models to retain object location information across entirely separate sessions, without the original image being re-provided. Under such conditions, none of the models tested would be capable of recalling spatial information, as all current LLMs reset completely between conversations with no persistent spatial knowledge retained.
This confirms the core limitation identified in Section 6: the complete absence of persistent spatial memory in current large language models.

5.5 Performance Summary

Scenario	GPT-4o	Gemini	Claude
Navigation	Good	Good	Best — Most Detailed
Object Interaction	Best — Practical Guidance	Partial — Avoided	Low — Declined
Scene Reasoning	Correct	Correct	Fail — Wrong Image
Spatial Memory	Context-Based Only	Context-Based Only	Context-Based Only

Key Takeaway: All three models demonstrated spatial understanding within a single conversation context, but none possess true persistent spatial memory. Performance varied significantly across tasks, with no single model consistently outperforming the others in all scenarios. Notably, GPT-4o excelled in practical object interaction guidance, Claude provided the most detailed navigation response, and all models failed to demonstrate genuine cross-session spatial memory.

5.6 Common Failure Patterns

Across all practical scenario tests, the following recurring failure patterns were identified in GPT-4o, Gemini, and Claude:

1. Universal Spatial Memory Failure No model tested was able to retain object positions between interactions. Each query requires fresh visual input, making continuous spatial tracking impossible across all three models.
2. Directional Relationship Errors All models demonstrate inconsistent handling of left/right and front/back relationships, particularly when the viewpoint or perspective changes between queries.
3. Depth and Occlusion Blindness Identifying occluded or partially hidden objects remains a consistent weakness. Models struggle to reason about objects that are behind, underneath, or otherwise not directly visible in the input image.
4. Confident Spatial Hallucination Models frequently generate spatially incorrect responses with high confidence, failing to acknowledge uncertainty in spatial descriptions or object positions.
5. Static Input Dependency All models rely exclusively on discrete, static image frames. None demonstrated the ability to adapt to dynamic or continuously changing environments in real time.
6. Spatial Context Confusion As observed in Scenario 3, models can incorrectly apply spatial analysis to the wrong image when multiple images are present in a conversation, leading to entirely inaccurate responses.

Key Takeaway: These failure patterns are not model-specific — they reflect fundamental architectural limitations shared across current LLMs, reinforcing the need for specialized spatial systems as outlined in Section 8.

6. Core Limitations Analysis

6.1 Fundamental Constraints

Based on research findings and benchmark testing, current LLMs exhibit the following core limitations:

1. No True 3D World Model

LLMs process 2D image representations, not genuine 3D spatial structures. They cannot:

- Reason about hidden objects behind visible ones
- Understand true depth relationships
- Predict 3D consequences of movements
- Maintain consistent 3D scene representations

2. Lack of Persistent Spatial Memory

Every interaction is independent:

- Cannot remember previous object positions
- No accumulation of spatial knowledge across sessions
- Cannot build long-term spatial maps
- Stateless processing of spatial queries

3. Weak Temporal Understanding

Limited ability to track changes over time:

- Cannot predict object movement trajectories accurately
- Struggles with "before and after" spatial comparisons
- Poor performance on dynamic scene analysis
- No true understanding of cause-and-effect in physical space

4. No Real-Time Grounding

Dependency on static input:

- Requires discrete image/video frames
- Cannot process continuous sensory streams
- Delayed feedback loops
- No direct connection to physical environment

5. Spatial Hallucination

Models frequently:

- Generate plausible but incorrect spatial descriptions
- Confuse left/right, front/back relationships
- Misestimate distances and sizes
- Provide confident but inaccurate spatial claims

6. Cannot Directly Act in Environment

Fundamental limitation:

- Output is text or discrete action tokens
- No direct motor control
- Requires translation layer to physical actions
- Cannot adjust based on proprioceptive feedback

7. Emerging Approaches and Research Directions

7.1 Vision-Language-Action (VLA) Models

Paradigm Shift:
VLA models represent a fundamental breakthrough by unifying perception, reasoning, and action in a single architecture.
Key Systems:
RT-2 (Robotic Transformer 2) - Google DeepMind:
- First true vision-language-action model
- Built on PaLM-E and PaLI-X backbones
- Co-trained on web-scale vision-language data and robot demonstrations
- Outputs discrete action tokens that control robots
- Demonstrates chain-of-thought reasoning for control
- Achieves generalization to new objects and environments

OpenVLA (Stanford University, UC Berkeley, Google DeepMind, Toyota Research Institute):
- 7B-parameter open-source VLA model
- Trained on 970k real-world robot demonstrations
- Fuses DINOv2 and SigLIP vision features with Llama-2 language backbone
- Outperforms RT-2 despite smaller size
- Available for public use and research

π0 (Pi-Zero) - Physical Intelligence:
- 3.3B-parameter model (3B PaliGemma VLM base + 300M action expert) adapted for real-time dexterous control
- Uses flow matching for continuous action outputs
- High-frequency motor command generation
- Supports complex manipulation tasks (laundry folding, egg handling)

Gemini Robotics:
- Vision-language-action model built on Gemini 2.0
- Physical actions as new output modality
- Direct robot control without intermediate systems

7.2 NeRF and 3D Scene Reconstruction

Neural Radiance Fields (NeRF):
Emerging technology for creating 3D spatial representations:
- Converts 2D images into 3D scene models
- Enables view synthesis and spatial understanding
- Potential integration with LLMs for true 3D reasoning

7.3 Spatial Memory Graphs and Scene Graphs

Scene Graph Approach:
Structured representations of spatial relationships:

- Nodes represent objects
- Edges represent spatial relationships
- Enables explicit reasoning about object arrangements
- Used in benchmarks like GQA-spatial

Spatial Memory Integration:
Future systems may incorporate:

- Persistent spatial databases
- Incremental scene graph updates
- Long-term spatial knowledge accumulation

7.4 Embodied AI and World Models

Embodied AI Systems:
Integration of AI models with physical or simulated bodies:

- Robots with VLA models
- Simulated agents in virtual environments
- Real-time sensorimotor learning

World Models:
Predictive systems that simulate:

- Physical dynamics
- Object interactions
- Spatial consequences of actions
- Future state predictions

Leading Research:
Companies and institutions developing embodied AI:

- Google DeepMind (Gemini Robotics)
- Physical Intelligence (π0 series)
- Stanford, UC Berkeley, Google DeepMind & Toyota Research Institute (OpenVLA)
- 1X Technologies (humanoid robots)

8. Role of LLM in Spatial AI Stack

8.1 What LLMs Are NOT

Critical Distinctions:
LLMs are NOT:

- Spatial engines (cannot compute precise spatial relationships)
- Perception systems (rely on pre-processed visual features)
- Motor control systems (cannot directly command actuators)
- 3D modeling engines (work with 2D representations)

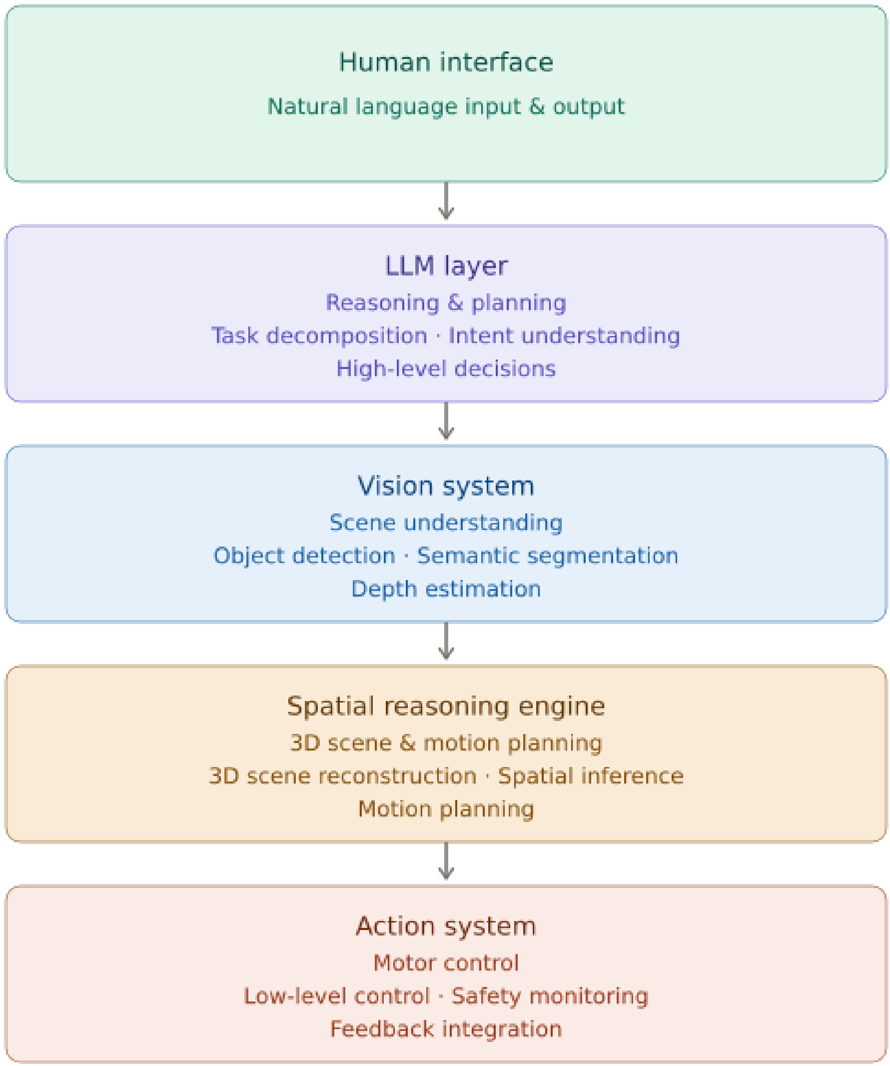
8.2 What LLMs ARE

Actual Capabilities:
LLMs ARE:

- Reasoning layers: High-level decision making and planning
- Planners: Breaking down complex tasks into steps
- Language interfaces: Translating human intent to machine actions
- Knowledge integrators: Combining semantic understanding with visual input

8.3 Spatial AI System Architecture

Proposed Stack:



Key Insight:
The most effective spatial AI systems use LLMs as the reasoning and planning layer, not as the entire solution. Specialized components handle perception, spatial computation, and motor control.

9. Key Findings and Implications

9.1 Can LLMs Understand Space or Just Describe It?

Answer: Current LLMs primarily describe space rather than truly understand it.
Evidence:

- High performance on semantic spatial descriptions
- Poor performance on precise spatial measurements
- Reliance on learned patterns rather than spatial models
- Frequent hallucinations in spatial details

9.2 Can They Reason About Movement and Interaction?

Answer: Limited capability, improving with VLA architectures.
Traditional LLMs:

- Can suggest movement strategies conceptually
- Cannot generate precise trajectories
- No direct understanding of physical constraints

VLA Models:

- Can generate actionable movement commands
- Learn from physical demonstrations
- Exhibit emergent spatial reasoning capabilities

9.3 What's Missing for Real-World Deployment?

Critical Gaps:

1. **Persistent Spatial Memory**
    - Need for long-term spatial maps
    - Cross-session knowledge retention
2. **True 3D Understanding**
    - Integration with 3D reconstruction systems
    - Native volumetric reasoning
3. **Real-Time Adaptation**
    - Continuous sensory processing
    - Dynamic environment tracking
4. **Robust Safety Mechanisms**
    - Physical constraint understanding
    - Collision avoidance
    - Force limitation
5. **Efficient Action Generation**
    - Low-latency control loops
    - High-frequency motor commands

9.4 What Needs to Be Combined with LLMs?

Essential Components:

Perception Systems:

- Advanced computer vision
- Depth sensors (LiDAR, stereo cameras)
- Object tracking systems

Spatial Processing:

- 3D reconstruction (NeRF, Gaussian Splatting)
- SLAM (Simultaneous Localization and Mapping)
- Scene graph generation

Action Generation:

- Motion planning algorithms
- Low-level controllers
- Safety monitoring systems

Memory Systems:

- Spatial databases
- Scene graph storage
- Episodic memory for spatial experiences

10. Implications for Spatial AI Assistants

10.1 Design Recommendations

For Developers Building Spatial AI Systems:

1. Don't rely on LLMs alone for spatial reasoning
2. Use LLMs as the reasoning layer in a multi-component architecture
3. Integrate specialized vision systems for perception
4. Implement dedicated spatial engines for 3D understanding
5. Consider VLA models for embodied AI applications
6. Build persistent spatial memory separate from LLM context

10.2 Current Best Practices

Model Selection:

- Use Gemini for applications requiring spatial understanding
- Use GPT-4o for multimodal reasoning with moderate spatial needs
- Use Claude for document analysis and interpretive tasks
- Use VLA models (RT-2, OpenVLA, π0) for robotic control

10.3 Future Outlook

Promising Trends:

1. **Convergence of LLMs and VLA models**
    - Single models handling reasoning and action
    - Native spatial understanding in foundation models
2. **Integration with 3D reconstruction**
    - Real-time scene understanding
    - Dynamic environment models
3. **Improved spatial memory architectures**
    - Persistent spatial knowledge bases
    - Cross-session spatial reasoning
4. **Enhanced embodied AI systems**
    - Humanoid robots with spatial intelligence
    - Autonomous navigation with natural language control

Conclusion

Spatial reasoning remains a significant challenge for current Large Language Models. While GPT-4o, Gemini, and Claude demonstrate impressive multimodal capabilities, true spatial intelligence requires integration with specialized systems. Gemini leads in spatial understanding, particularly through dedicated robotics variants, while Claude explicitly acknowledges spatial limitations. The emergence of Vision-Language-Action models represents the most promising path forward, unifying perception, reasoning, and action in single architectures. For real-world deployment of Spatial AI Assistants, developers must combine LLMs with vision systems, spatial reasoning engines, and action generation components, treating LLMs as the reasoning layer rather than the complete solution.

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Source: Google DeepMind - https://deepmind.google/blog/rt-2-new-model-translates-vision-and-language-into-action/
Official announcement of RT-2 vision-language-action model

17. Gemini 3 for developers: New reasoning, agentic capabilities (November 2025)
Source: Google Developers Blog - https://blog.google/innovation-and-ai/technology/developers-tools/gemini-3-developers/
Developer-focused overview of Gemini 3's spatial reasoning for embodied tasks

18. Planet and Anthropic Partner to Use Claude for Geospatial Analysis (March 2025)
Source: Business Wire - https://www.businesswire.com/news/home/20250306606139/en/
Application of Claude's visual reasoning to satellite imagery analysis